

Dynamic Grouped-Query Attention: Predictive, Input-Dependent Key-Value Head Sharing via Learned Collaboration Templates

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Abstract

Grouped-Query Attention (GQA) shrinks the KV cache of Transformer language models by statically sharing key/value heads. We ask whether sharing can instead be decided dynamically per input, formalize the resulting architecture (Dynamic GQA), and prove its cache and FLOP savings, a cache-consistency guarantee under causal decoding, and a spectral bound on the output error that KV sharing introduces. We then test the idea’s central premise with a pre-registered, training-free protocol on three model families (Pythia-2.8B, Qwen3-4B, Nemotron-Mini-4B), run end-to-end on rented cloud GPUs (RTX 3090/4090 and an NVIDIA A100) for under US\$5. Three results emerge. First, the dynamic premise fails as stated: raw KV-head activations are nearly orthogonal (mean cosine 0.002) and their similarity structure barely varies across domains (adjusted Rand index 0.59–0.86), so a token-level router has little to exploit. Second, reverse-engineering that failure shows the redundancy is real but hidden by a change of basis: after fitting closed-form linear maps between head pairs, similarity rises to 0.89–0.96 with 96–100% of pairs above 0.80, independently corroborating concurrently reported inter-head linear structure. Third, exploiting this structure preserves the full H/G cache reduction at a few percent FLOP overhead and beats naive merging in every one of our 56 comparisons, by up to three orders of magnitude in perplexity; in a controlled benchmark of training-free conversions at matched cache budget, reconstruction quality rises with the information given to the reconstructor, and the multi-reference variant reaches +17–20% perplexity at $2\times$ compression beyond Qwen3-4B’s built-in GQA, with no gradient step taken. The evidence redirects the design space: similarity-aware grouping with per-head linear corrections, not per-token clustering of raw heads.

1 Introduction

The memory footprint of the key-value (KV) cache is a dominant cost of autoregressive inference in large language models (LLMs). For a model with L layers, H attention heads of dimension d_h , batch size B , and context length n , the cache stores $2 L H d_h B n$ activations, which for long contexts routinely exceeds the memory occupied by the model weights themselves. Multi-Query Attention (MQA) (Shazeer, 2019) and Grouped-Query Attention (GQA) (Ainslie et al., 2023) attack this cost by letting several query heads share a single key and value head, shrinking the cache by a factor of H/G where G is the number of KV groups. GQA is now standard in open-weight models such as Llama and Qwen.

GQA’s grouping, however, is *static*: the assignment of query heads to KV groups is fixed at training (or “uptraining”) time and applied identically to every token of every input. This is at odds with a well-documented empirical fact: the functional roles of attention heads, and hence the redundancy between them, vary across inputs and domains (Michel et al., 2019; Voita et al., 2019). A pair of heads that is interchangeable on source code may be sharply differentiated on poetry. A static partition must therefore be conservative: it can only exploit redundancy that holds on average over the entire training distribution.

This paper develops the following hypothesis:

A lightweight predictor can determine, from the hidden state alone and before keys and values are computed, which attention heads will behave similarly on the current input; letting exactly those heads share KV computation preserves model quality while reducing both projection compute and KV-cache memory beyond what a static grouping of equal size achieves.

Two design commitments distinguish the proposal from prior adaptive-GQA work (Section 2). First, grouping is *predictive*: the router acts on the hidden state x_t *before* the KV projections are applied, so redundant keys and values are never computed. Approaches that first compute all per-head keys and then merge or re-allocate them (e.g., key-norm-driven allocation (Khan et al., 2024)) compress representations post hoc and cannot save the projection FLOPs. Second, grouping is *input-dependent at inference time* but constrained to a small learned set of *collaboration templates*, which (i) makes routing a cheap M -way classification rather than an online clustering problem, and (ii) is exactly the structure needed to keep the KV cache consistent under autoregressive decoding (Section 4.3).

Contributions. Concretely: (1) we give a unified formal treatment of MHA, MQA, and GQA as special cases of a group map, and define Dynamic GQA as routing over a family of such maps (Section 4); (2) we prove exact KV-cache and projection-FLOP savings (Propositions 5.1 and 5.2), a spectral perturbation bound on the attention output under KV sharing (Theorem 5.4) with a lossless-sharing corollary (Corollary 5.5), and a layer-level error bound (Corollary 5.6); (3) we formalize the cache-consistency requirement for dynamic grouping under causal decoding and prove that template routing satisfies it (Proposition 4.6); (4) we specify a low-cost, training-free empirical protocol with pre-registered decision criteria, and *execute* it on Qwen3-4B and Pythia-2.8B, reporting both the negative verdicts on the strong hypothesis and a robust positive finding on similarity-aware grouping (Sections 6–7).

2 Related Work

Attention and static KV sharing. Multi-head attention was introduced by Vaswani et al. (2017). MQA (Shazeer, 2019) shares a single KV head across all query heads; GQA (Ainslie et al., 2023) interpolates between MHA and MQA with G static groups and shows that existing MHA checkpoints can be “uptrained” into GQA models cheaply. Multi-head Latent Attention (MLA) (DeepSeek-AI, 2024) compresses keys and values into a shared low-rank latent, reducing the cache by over 90%; like GQA, the compression structure is fixed across inputs.

Adaptive and non-uniform grouping. Several works relax GQA’s uniform grouping while remaining static at inference time. QCQA (Joshi et al., 2024) searches for quality-optimal head groupings with an evolutionary algorithm; WGQA (Chinnakonduru & Mohapatra, 2024) learns weighted (rather than mean-pooled) KV aggregation during fine-tuning; DHA (Chen et al., 2024) fuses similar heads layer-wise when converting MHA checkpoints. KDGQA and its dynamic variant DGQA (Khan et al., 2024) allocate query heads to key groups using evolving key-norm statistics; the allocation adapts during *training*, and the key projections are still all computed. Our proposal differs on both axes: grouping is decided per-input at *inference* time, and it is decided *before* the KV projections are executed, so the saving is in computation and cache rather than in re-allocation of already-computed heads.

Head redundancy. Michel et al. (2019) showed that a large fraction of attention heads can be pruned post-training with little quality loss, and Voita et al. (2019) identified specialized, partially interchangeable head roles. These results establish that head redundancy exists on average; our hypothesis is the sharper claim that redundancy has predictable *input-dependent structure*.

Head alignment and linear structure. Closest to our Section 7.3: CHAI (Agarwal et al., 2024) clusters correlated heads at runtime to share attention computation; Jin et al. (2024) apply Procrustes alignment to attention heads before merging them for MHA→GQA conversion; and concurrent work (Shaikh et al., 2026) documents pervasive inter-head linear predictability of Q/K/V activations at scale and exploits it by caching reference heads and reconstructing the rest through linear maps. Our alignment experiment independently

arrives at the same structural phenomenon via a different route (a pre-registered test of dynamic grouping that failed on raw activations), and adds a controlled raw-vs-aligned comparison at fixed cache budget and clustering. Our input-dependence analysis (Finding 2) also qualifies CHAI-style *runtime* clustering: we find the cluster structure to be largely static across domains, suggesting offline clustering suffices.

Conditional computation. Routing tokens to a sparse subset of parameters is well established for feed-forward layers via mixtures of experts (Shazeer et al., 2017; Fedus et al., 2022). Dynamic GQA can be read as MoE-style routing applied to the *KV projections* of attention, with the crucial difference that experts here are *partitions* (templates) rather than independent subnetworks, which is what preserves cache consistency (Section 4.3). KV-cache eviction methods such as H₂O (Zhang et al., 2023) are orthogonal: they drop cached entries along the sequence axis, whereas we share them along the head axis; the two compose.

3 Preliminaries

3.1 Notation

For a matrix A , $\|A\|_2$ denotes the spectral norm (largest singular value) and $\|a\|_2$ the Euclidean norm of a vector. Δ^{k-1} is the probability simplex in $\mathbb{R}^k$. $[H] := \{1, \dots, H\}$. Row vectors of a matrix $A \in \mathbb{R}^{n \times d}$ are written $a_i \in \mathbb{R}^d$. The softmax over a vector $z \in \mathbb{R}^n$ is $\text{softmax}(z)_i = e^{z_i} / \sum_j e^{z_j}$; applied to a matrix it acts row-wise.

3.2 Attention as a group map

Let $X \in \mathbb{R}^{n \times d}$ be the layer input for a sequence of n tokens with model dimension d , and let there be H query heads of dimension d_h (typically $d_h = d/H$).

Definition 3.1 (Grouped attention). *Fix a number of groups $G \in [H]$ and a surjective group map $g : [H] \rightarrow [G]$. Grouped attention with parameters $\{W_Q^h\}_{h=1}^H, \{W_K^j, W_V^j\}_{j=1}^G \subset \mathbb{R}^{d \times d_h}$ and $W_O \in \mathbb{R}^{H d_h \times d}$ computes, for each head $h \in [H]$,*

$$Q_h = XW_Q^h, \quad K_{g(h)} = XW_K^{g(h)}, \quad V_{g(h)} = XW_V^{g(h)}, \quad (1)$$

$$O_h = \text{softmax}\left(\frac{1}{\sqrt{d_h}} Q_h K_{g(h)}^\top + M\right) V_{g(h)}, \quad (2)$$

where M is the (causal) attention mask, and outputs $\text{Concat}(O_1, \dots, O_H) W_O$.

Remark 3.2. Definition 3.1 unifies the standard mechanisms: $G = H$ with $g = \text{id}$ recovers MHA (Vaswani et al., 2017); $G = 1$ recovers MQA (Shazeer, 2019); and $1 < G < H$ with the uniform map $g(h) = \lceil hG/H \rceil$ recovers GQA (Ainslie et al., 2023). In all cases the group map is a single, fixed g applied to every input. Positional encodings such as RoPE act on Q_h and $K_{g(h)}$ after projection and are independent of the grouping; we suppress them for clarity.

3.3 Baseline costs

Throughout, “per token, per layer” costs refer to autoregressive decoding of one new token. Under Definition 3.1: the KV cache stores $2Gd_h$ scalars per token per layer; the Q/K/V projections cost $2d d_h(H + 2G)$ FLOPs per token (counting a multiply-add as two FLOPs); and the attention-score and value-mixing computation costs $O(n H d_h)$ per token *regardless of G* , since every query head still forms scores against its group’s keys. KV sharing therefore reduces projection compute and cache traffic, not score computation, a property Dynamic GQA inherits.

4 Dynamic Grouped-Query Attention

4.1 Motivation: why not merge computed KVs?

A natural first idea is to compute all per-head keys $K_1, \dots, K_H$, detect similar ones, and merge them, e.g. $K_{\text{avg}} = \frac{1}{2}(K_1 + K_2)$. This is doubly deficient. First, it saves no projection compute, every key has already

been computed. Second, distinct heads learn distinct projection subspaces: $K_h = XW_K^h$, and for generic $W_K^1 \neq W_K^2$ the average $\frac{1}{2}X(W_K^1 + W_K^2)$ lies in neither head’s learned subspace, so the merged key preserves the semantics of neither head (cf. the head-fusion difficulties documented by [Chen et al. \(2024\)](#)). The correct formulation is to decide the sharing structure *before* computing keys and values, and to give each shared group its own learned projection.

4.2 Collaboration templates and routing

Definition 4.1 (Template family). *A template family of size M is a set $\mathcal{T} = \{g^{(1)}, \dots, g^{(M)}\}$ of group maps $g^{(m)} : [H] \rightarrow [G_m]$, together with per-template KV projections $\{W_K^{(m,j)}, W_V^{(m,j)} : j \in [G_m]\}$. Query projections $\{W_Q^h\}_{h=1}^H$ and the output projection W_O are shared across templates.*

The templates play the role of “collaboration patterns”: e.g., one template may group heads $\{2, 5, 9\}$ that specialize jointly on source code, another may group $\{3, 7, 8\}$ for mathematical text. Templates are learned, not hand-designed (Section 4.4).

Definition 4.2 (Router). *A router is a map $R : \mathbb{R}^d \rightarrow \Delta^{M-1}$. Given a routing representation $\bar{x} \in \mathbb{R}^d$ (Definition 4.3), the selected template is $m^* = \arg \max_m R(\bar{x})_m$. We take R to be a linear-softmax classifier, $R(\bar{x}) = \text{softmax}(W_R \bar{x})$ with $W_R \in \mathbb{R}^{M \times d}$, optionally preceded by a low-rank bottleneck.*

Definition 4.3 (Routing granularity). *Routing may be applied at three granularities, with $\bar{x}$ respectively: sequence-level ($\bar{x} = \frac{1}{n} \sum_t x_t$ over a prefix, one template per sequence), segment-level ($\bar{x}$ pooled over a chunk of c tokens, one template per chunk), or token-level ($\bar{x} = x_t$, one template per token).*

Definition 4.4 (Dynamic GQA layer). *Given a template family $\mathcal{T}$ and router R , the Dynamic GQA layer computes, for each token t with selected template m_t ,*

$$q_{t,h} = W_Q^h \top x_t, \quad k_{t,j} = W_K^{(m_t,j)} \top x_t, \quad v_{t,j} = W_V^{(m_t,j)} \top x_t, \quad j \in [G_{m_t}], \quad (3)$$

caches $(m_t, \{(k_{t,j}, v_{t,j})\}_{j \in [G_{m_t}]})$, and for each head h forms attention over the per-position keys

$$\kappa_h(s) := k_{s, g^{(m_s)}(h)}, \quad \nu_h(s) := v_{s, g^{(m_s)}(h)}, \quad s \leq t, \quad (4)$$

i.e. $O_{t,h} = \sum_{s \leq t} a_{t,h}(s) \nu_h(s)$ with $a_{t,h} = \text{softmax}(\langle q_{t,h}, \kappa_h(\cdot) \rangle / \sqrt{d_h})$.

Setting $M = 1$ recovers GQA exactly, so Dynamic GQA is a strict generalization.

4.3 Cache consistency under causal decoding

A subtlety absent from the static setting deserves emphasis. Under causal attention, the key that head h uses for a *past* position s was committed to the cache at time s , under whatever grouping was active at time s . Any dynamic scheme must therefore satisfy:

Definition 4.5 (Cache consistency). *A dynamic grouping scheme is cache-consistent if, for every head h and every pair of positions $s \leq t$, the key/value that head h reads at query time t for position s is a deterministic function of information already stored at time s , without recomputation.*

Proposition 4.6 (Template routing is cache-consistent). *The Dynamic GQA layer of Definition 4.4 is cache-consistent, with per-position cache cost $2G_{m_s}d_h + O(1)$ scalars (the $O(1)$ term storing the template index m_s).*

Proof. At time s the layer stores m_s and the G_{m_s} shared key/value pairs. At any later time t , head h ’s key for position s is $\kappa_h(s) = k_{s, g^{(m_s)}(h)}$ by (4), which depends only on the stored pairs and the stored index m_s ; the map $h \mapsto g^{(m_s)}(h)$ is a fixed lookup into the (static, model-level) template family. No projection is recomputed. $\square$

Remark 4.7. *Proposition 4.6 is precisely why routing over a finite template family is preferable to unconstrained per-token clustering: an arbitrary clustering at time t of keys that were never materialized per-head*

at time $s < t$ is not well defined, and materializing them defeats the purpose. The price of token-level routing is a ragged cache layout (different positions store different numbers of KV vectors) and a per-position index gather; segment- or sequence-level routing (Definition 4.3) restores a rectangular layout within each segment and is the recommended starting point.

4.4 Training and differentiability

The arg max in Definition 4.2 is non-differentiable. Standard remedies apply: Gumbel-softmax relaxation with straight-through estimation (Jang et al., 2017) during training and hard routing at inference, or an auxiliary load-balancing loss as in switch-style MoE (Fedus et al., 2022) to prevent template collapse:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda_{\text{bal}} M \sum_{m=1}^M f_m \bar{p}_m + \lambda_{\text{cons}} \mathbb{E}_t [1 - \langle R(\bar{x}_t), R(\bar{x}_{t-1}) \rangle], \quad (5)$$

where f_m is the fraction of routing decisions sent to template m , $\bar{p}_m$ the mean router probability of template m , and the last term (optional) encourages temporally stable routing, which reduces raggedness of the cache under token-level routing. Templates can be initialized from an uptrained GQA checkpoint (Ainslie et al., 2023): initialize every $W_K^{(m,j)}$ by mean-pooling the MHA key projections of the heads in group j of template m (and analogously for values), then recover quality with parameter-efficient fine-tuning (Hu et al., 2022; Dettmers et al., 2023).

5 Theoretical Analysis

We now quantify (i) what Dynamic GQA saves, and (ii) what it can cost in fidelity. All results are per layer; summation over layers is immediate.

5.1 Memory and compute

Proposition 5.1 (KV-cache size). *Let b bytes be the storage precision. Under Dynamic GQA with routing distribution $\pi \in \Delta^{M-1}$ over templates (i.e., π_m is the long-run fraction of positions routed to template m), the expected KV-cache cost per token per layer is*

$$C_{\text{DGQA}} = 2b d_h \mathbb{E}_{m \sim \pi} [G_m] + O(1), \quad (6)$$

versus $C_{\text{MHA}} = 2b d_h H$ and $C_{\text{GQA}} = 2b d_h G$. The compression ratio relative to MHA is $H/\mathbb{E}_{\pi} [G_m]$ (up to the $O(1)$ index term).

Proof. By Proposition 4.6, a position routed to template m stores G_m key vectors and G_m value vectors of dimension d_h at precision b , plus the index m_s ($\lceil \log_2 M \rceil$ bits). Taking expectation over $m \sim \pi$ gives the claim; MHA and GQA are the special cases $\pi = \delta_{g=\text{id}}$ and $\pi = \delta_{g(1)}$, $M = 1$. $\square$

Proposition 5.2 (Projection FLOPs and router overhead). *Per token per layer, counting a multiply-add as 2 FLOPs, Dynamic GQA costs*

$$F_{\text{DGQA}} = \underbrace{2d d_h H}_{Q \text{ proj.}} + \underbrace{4d d_h \mathbb{E}_{\pi} [G_m]}_{K, V \text{ proj.}} + \underbrace{2dM}_{\text{router}}, \quad (7)$$

versus $F_{\text{MHA}} = 2d d_h H + 4d d_h H = 6d^2$ when $d_h = d/H$. The router overhead relative to the MHA projection cost is therefore $\frac{2dM}{6d^2} = \frac{M}{3d}$, i.e. negligible whenever $M \ll d$ (for $d = 2048$, $M = 16$: $\approx 0.26\%$). Attention-score computation is unchanged.

Proof. Each of the H query projections is a $d \times d_h$ matrix-vector product ($2d d_h$ FLOPs). A position routed to template m computes G_m key and G_m value projections, giving $4d d_h G_m$ in expectation $4d d_h \mathbb{E}_{\pi} [G_m]$. The linear router costs one $M \times d$ matrix-vector product, $2dM$ FLOPs. The MHA total is $6d d_h H = 6d^2$ when $d_h = d/H$; the stated ratio follows. $\square$

5.2 Fidelity: a perturbation bound for KV sharing

The substantive question is how much a head’s output changes when its private keys and values are replaced by shared ones. Fix a head h , write $K := K_h$, $V := V_h \in \mathbb{R}^{n \times d_h}$ for the keys and values head h *would have computed privately* (its MHA counterfactual), and $\tilde{K}, \tilde{V}$ for the shared matrices assigned by the active template. We bound the change in the attention output of an arbitrary query row $q := q_i \in \mathbb{R}^{d_h}$.

We use the following Lipschitz property of the softmax (Gao & Pavel, 2017).

Lemma 5.3 (Softmax is $\frac{1}{2}$ -Lipschitz). *For all $z, z' \in \mathbb{R}^n$, $\|\text{softmax}(z) - \text{softmax}(z')\|_2 \leq \frac{1}{2} \|z - z'\|_2$.*

Proof. The Jacobian of the softmax at z is $J(z) = \text{diag}(p) - pp^\top$ with $p = \text{softmax}(z)$, which is the covariance matrix of the categorical one-hot random vector with distribution p ; hence $J(z) \succeq 0$. Its largest eigenvalue is bounded: for any unit u , $u^\top J u = \sum_i p_i u_i^2 - (\sum_i p_i u_i)^2 = \text{Var}_p(u) \leq \frac{1}{4} (\max_i u_i - \min_i u_i)^2 \leq \frac{1}{4} \cdot (2 \|u\|_\infty)^2 \leq 1$, where the variance bound is Popoviciu’s inequality. A sharper constant follows from the refined analysis of Gao & Pavel (2017), who show $\|J(z)\|_2 \leq \frac{1}{2}$ for all z ; the claim then follows from the mean-value inequality $\|\text{softmax}(z) - \text{softmax}(z')\|_2 \leq \sup_\xi \|J(\xi)\|_2 \|z - z'\|_2$. $\square$

Theorem 5.4 (Output perturbation under KV sharing). *Let $a = \text{softmax}(Kq/\sqrt{d_h})$ and $\tilde{a} = \text{softmax}(\tilde{K}q/\sqrt{d_h})$ (with the same causal mask), and let $o = V^\top a$, $\tilde{o} = \tilde{V}^\top \tilde{a}$ be the head’s output for query q under private and shared KV respectively. Then*

$$\|o - \tilde{o}\|_2 \leq \underbrace{\frac{\|V\|_2 \|q\|_2}{2\sqrt{d_h}} \|K - \tilde{K}\|_2}_{\text{score distortion}} + \underbrace{\|V - \tilde{V}\|_2}_{\text{value distortion}}. \quad (8)$$

Proof. Decompose $o - \tilde{o} = V^\top (a - \tilde{a}) + (V - \tilde{V})^\top \tilde{a}$, so by the triangle inequality

$$\|o - \tilde{o}\|_2 \leq \|V\|_2 \|a - \tilde{a}\|_2 + \|V - \tilde{V}\|_2 \|\tilde{a}\|_2. \quad (9)$$

For the first term, the logit vectors are $z = Kq/\sqrt{d_h}$ and $\tilde{z} = \tilde{K}q/\sqrt{d_h}$, so $\|z - \tilde{z}\|_2 = \|(K - \tilde{K})q\|_2 / \sqrt{d_h} \leq \|K - \tilde{K}\|_2 \|q\|_2 / \sqrt{d_h}$, and Lemma 5.3 gives $\|a - \tilde{a}\|_2 \leq \|K - \tilde{K}\|_2 \|q\|_2 / (2\sqrt{d_h})$. (Masked positions receive logit $-\infty$ in both z and $\tilde{z}$ and contribute zero probability in both, so the bound applies to the unmasked sub-vector.) For the second term, $\tilde{a}$ is a probability vector, hence $\|\tilde{a}\|_2 \leq \|\tilde{a}\|_1 = 1$. Substituting into (9) yields (8). $\square$

Corollary 5.5 (Lossless sharing under exact redundancy). *If the heads grouped by the active template are exactly redundant, i.e. $K_h = \tilde{K}$ and $V_h = \tilde{V}$ for every head h in the group, then Dynamic GQA reproduces the MHA output exactly for those heads.*

Proof. Both terms of (8) vanish, so O_h in (2) is unchanged for every grouped head; the concatenation and output projection are shared, so the layer output is identical. $\square$

Corollary 5.6 (Layer-level bound). *Let δ_h denote the right-hand side of (8) for head h (with $\delta_h = 0$ for ungrouped heads), and let $y_i, \tilde{y}_i$ be the attention-layer outputs for token i under private and shared KV. Then*

$$\|y_i - \tilde{y}_i\|_2 \leq \|W_O\|_2 \left(\sum_{h=1}^H \delta_h^2 \right)^{1/2}. \quad (10)$$

Proof. The concatenated per-head error vector $e = (o_1 - \tilde{o}_1, \dots, o_H - \tilde{o}_H) \in \mathbb{R}^{H d_h}$ has $\|e\|_2 = (\sum_h \|o_h - \tilde{o}_h\|_2^2)^{1/2} \leq (\sum_h \delta_h^2)^{1/2}$, and $y_i - \tilde{y}_i = W_O^\top e$, so $\|y_i - \tilde{y}_i\|_2 \leq \|W_O\|_2 \|e\|_2$. $\square$

Lemma 5.7 (Weight-space control). *With $K = XW_K^h$ and $\tilde{K} = X\tilde{W}_K$,*

$$\|K - \tilde{K}\|_2 \leq \|X\|_2 \|W_K^h - \tilde{W}_K\|_2, \quad (11)$$

and analogously for values.

Proof. Submultiplicativity of the spectral norm: $\|X(W_K^h - \tilde{W}_K)\|_2 \leq \|X\|_2 \|W_K^h - \tilde{W}_K\|_2$. $\square$

Remark 5.8 (What the theory does and does not establish). *Theorem 5.4 and Lemma 5.7 establish that KV sharing is safe when the grouped heads are close, in the input-dependent quantities $\|K_h - \tilde{K}\|_2$, $\|V_h - \tilde{V}\|_2$, which are themselves controlled by weight-space proximity scaled by the input energy $\|X\|_2$. This is a soundness guarantee, not a proof that useful redundancy exists: whether real models exhibit head clusters tight enough for the bound to be small, and whether cluster membership is predictable from $\bar{x}$ alone, are empirical questions. The protocol of Section 6 is designed to answer exactly these questions before any architecture is trained. The bound is also worst-case over query directions; average-case distortion can be far smaller when $(K - \tilde{K})q$ concentrates in low-attention regions.*

Remark 5.9 (Comparison with static GQA). *For static GQA, (8) holds with $\tilde{K}$ fixed per group across all inputs, so the relevant quantity is $\mathbb{E}_X[\|K_h(X) - \tilde{K}(X)\|_2]$ averaged over the whole data distribution. Dynamic GQA replaces this with a conditional quantity: each input is routed to the template minimizing distortion for that input. Whenever head similarity structure varies across the input distribution, $\min_m \mathbb{E}_X[\cdot \mid m^*(X) = m] \leq \mathbb{E}_X[\cdot]$ pointwise, so the achievable distortion at equal $\mathbb{E}[G]$ is no worse, and strictly better under non-trivial variation. This is the formal version of the paper’s central intuition.*

6 Empirical Validation Protocol

The hypothesis is falsifiable without training any model. We propose a staged protocol on pretrained open-weight models (Llama-3.2-3B, Qwen3-4B), executable on a single rented consumer GPU (e.g. RTX 4090 class) for approximately US\$10–30.

Stage 1: Activation collection (inference only). Register forward hooks on every attention module and record per-head Q_h, K_h, V_h and head outputs O_h for 5,000–20,000 tokens drawn from balanced domains (code, mathematics, English prose, poetry). No gradients are computed.

Stage 2: Redundancy measurement (offline, CPU). For each layer ℓ and token t , compute pairwise similarities between heads: cosine similarity of key/value vectors, and Jensen–Shannon divergence between attention distributions. Define the *redundancy profile*

$$\rho_\ell(\tau) = \frac{2}{H(H-1)} \sum_{h < h'} \mathbf{1}[\text{sim}_\ell(h, h') > \tau], \quad (12)$$

and measure: (i) the magnitude of $\rho_\ell(\tau)$ at, e.g., $\tau = 0.9$; (ii) its stability across layers; and crucially (iii) whether the *cluster structure* (from agglomerative or spectral clustering of the similarity matrices) differs significantly across domains. Outcome (iii) is the discriminating measurement: high redundancy with input-*invariant* clusters argues for better static GQA (or DHA-style fusion), not for dynamic routing; input-*dependent* clusters are the evidence Dynamic GQA needs.

Stage 3: Oracle simulation (no new kernels). Simulate Dynamic GQA with an *oracle router*: for each evaluation segment, cluster heads using Stage-2 statistics from held-out data of the same domain, then overwrite each grouped head’s K_h, V_h tensors in the forward pass with the group mean (or the group medoid, avoiding the subspace objection of Section 4 at simulation time). Report Δ -perplexity, next-token top-1 agreement with the base model, and mean KL divergence of logits, as functions of the effective compression $H/\mathbb{E}[G]$, against static GQA groupings of equal $\mathbb{E}[G]$ simulated identically. The oracle gap between dynamic and static curves upper-bounds what any learned router can gain.

Stage 4: Learned router and recovery (conditional). Only if the oracle gap is material: train the linear router of Definition 4.2 to predict oracle cluster assignments from pooled hidden states (a cheap supervised probe; its accuracy directly tests the “predictable from $\bar{x}$ alone” clause of the hypothesis), then recover quality with LoRA/QLoRA (Hu et al., 2022; Dettmers et al., 2023) on the 3–4B model under objective (5).

Model	Quantity	prose	math	code	stories
Qwen3-4B	mean off-diag. K-cosine	0.0024	0.0024	0.0023	0.0025
Qwen3-4B	max off-diag. K-cosine	0.42	0.45	0.40	0.46
Qwen3-4B	mean off-diag. V-cosine	−0.0001	0.0003	−0.0003	0.0006
Qwen3-4B	$\rho_\ell(0.80)$ (all layers)	0.00	0.00	0.00	0.00
Qwen3-4B	min pairwise query-head JS	0.007	0.010	0.007	0.007
Pythia-2.8B	$\rho_\ell(0.90)$ (all layers)	0.00	0.00	0.00	0.00

Table 1: Stage 2 redundancy measurements. KV-head activations are nearly orthogonal on average in both models; no head pair in any layer or domain exceeds cosine similarity 0.80. In contrast, some *query* heads have nearly duplicate attention distributions (minimum pairwise Jensen–Shannon divergence ≈ 0.007).

Decision criteria. Pre-registered thresholds avoid post-hoc rationalization. Suggested: proceed past Stage 2 if $\geq 30\%$ of head pairs exceed $\tau = 0.9$ similarity in at least a quarter of layers *and* cluster structure differs across domains with adjusted Rand index < 0.5 ; proceed past Stage 3 if dynamic oracle grouping achieves $\leq 1\%$ perplexity degradation at a compression where equal- $\mathbb{E}[G]$ static grouping degrades by $\geq 2\times$ that amount.

7 Results

We executed Stages 1–3 on two pretrained models: **Qwen3-4B** (GQA: $L=36$, $H_q=32$, $H_{kv}=8$, $d_h=128$; 8,192 tokens per domain) and **Pythia-2.8B** (MHA: $L=32$, $H=32$, $d_h=80$; 6,144 tokens per domain), on four domains (Wikipedia prose, GSM8K mathematics, MBPP Python code, TinyStories narrative), with sequence length 512. Stages 1–3 ran on a single rented RTX 3090 (24 GB); the alignment and comparison experiments of Sections 7.3–7.4 ran on an NVIDIA A100-SXM (80 GB). Total compute cost was **under US\$1**. For the already-grouped Qwen3-4B, the experiment tests whether *further* input-dependent grouping of its 8 KV heads is possible; Pythia-2.8B tests the hypothesis in its original MHA setting. The “dynamic” condition uses an oracle router (per-domain agglomerative clustering of the mean key-similarity matrices, fitted on held-out collection data), which upper-bounds any learned router; sharing is simulated by replacing grouped heads’ key/value activations with the group mean, i.e. *without* learned shared projections or any fine-tuning. The measurement and simulation code, run logs, and complete result CSVs accompany this paper as supplementary material.

7.1 Stage 2: redundancy and cluster stability

Table 1 summarizes Stage 2. Both pre-registered Stage-2 criteria *fail*, decisively:

Finding 1 (no activation-level KV redundancy). In both models, mean pairwise cosine similarity between distinct KV heads’ activations is ≈ 0.002 , the maximum observed pair is 0.46, and the redundancy profile $\rho_\ell(\tau)$ is exactly zero at $\tau = 0.80$ for every layer and domain. Distinct heads occupy nearly orthogonal subspaces. Notably this holds for MHA Pythia-2.8B as much as for GQA Qwen3-4B, so it is not merely an artifact of GQA uptraining having already removed redundancy.

Finding 2 (similarity structure is input-independent). Cross-domain adjusted Rand index between per-domain head clusterings is 0.59 ($G=2$) and 0.63 ($G=4$) on Qwen3-4B and 0.82–0.86 on Pythia-2.8B, far above the pre-registered < 0.5 threshold. Whatever weak similarity structure exists is largely a property of the *model*, not of the input. This directly undercuts the premise of *token-level* dynamic routing: an input-conditioned router has little input-dependent signal to exploit at the level of existing heads.

Finding 3 (query-side redundancy exists). The minimum pairwise Jensen–Shannon divergence between query heads’ attention distributions on Qwen3-4B is ≈ 0.007 , some query heads attend almost identically, echoing Michel et al. (2019); Voita et al. (2019). Redundancy in modern models appears to live in *where heads look* (attention patterns), not in *what they write* (K/V subspaces).

Domain	G	Compression	PPL (static)	PPL (dynamic)
<i>Pythia-2.8B (MHA, $H=32$); baseline PPL: prose 10.1, math 6.4, code 3.7, stories 5.1</i>				
prose	16	2×	665	375
math	16	2×	456	215
code	16	2×	506	112
stories	16	2×	170	92
code	8	4×	1652	1213
stories	8	4×	279	234
<i>Qwen3-4B (GQA, $H_{kv}=8$); baseline PPL: prose 13.6, math 3.6, code 4.3, stories 5.1</i>				
math	2	4×	16044	3552
code	2	4×	81700	20669
prose	2	4×	45412	63855
stories	2	4×	11646	26682

Table 2: Stage 3: perplexity under training-free KV-head activation merging, uniform static grouping vs. similarity-clustered (“dynamic oracle”) grouping at equal group count. Bold marks the better condition. Selected rows shown; over all 20 comparisons (12 Pythia, 8 Qwen), similarity-clustered grouping wins 11 of 12 on Pythia (all except math $G=4$) by up to 4.5× (code, $G=16$), while on already-grouped Qwen it wins only the two highest-compression cases shown. *All* conditions degrade catastrophically relative to baseline, as Theorem 5.4 predicts given Finding 1.

7.2 Stage 3: oracle sharing simulation

Table 2 reports the simulation. The pre-registered Stage-3 success criterion ($\leq 1\%$ dynamic degradation) fails by orders of magnitude, and this is exactly what the theory predicts: by Theorem 5.4 the output error is controlled by $\|K_h - \tilde{K}\|_2$, and Finding 1 shows these distances are large: there are no close head pairs, so training-free activation merging cannot be cheap. The quantitative agreement between the measured orthogonality and the observed catastrophic degradation is an empirical confirmation of the perturbation analysis (and of the subspace argument of Section 4).

Finding 4 (similarity-aware grouping dominates uniform grouping in MHA). Within the degraded regime, the grouping *choice* matters enormously in the MHA setting: similarity-clustered grouping beats GQA-style uniform contiguous grouping in 11 of 12 Pythia comparisons, with the gap *widening* as compression increases, up to 4.5× lower perplexity (code, $G=16$). On the already-grouped Qwen3-4B the picture is mixed (2 of 8, though those two wins are the largest in the study: 4.5× on math and 4.0× on code at $G=2$): the model’s uptrained KV heads leave little clustering structure to exploit, consistent with Finding 1. Since Finding 2 shows the clusterings are largely domain-stable, the Pythia benefit is attributable to *similarity-aware grouping* rather than to input-dependence per se.

7.3 Reverse-engineering the failure: alignment restores similarity

Finding 1 admits two readings: heads may be truly non-redundant, or raw cosine may be the wrong instrument, each head writes in its own learned basis, so two *functionally* similar heads can produce nearly orthogonal activation vectors. To discriminate, we fit, for every ordered head pair (g, h) in every layer of Pythia-2.8B, a ridge least-squares map

$$R_{g \rightarrow h} = \arg \min_R \|K_g R - K_h\|_F^2 + \lambda \|R\|_F^2 = (K_g^\top K_g + \lambda I)^{-1} K_g^\top K_h, \quad (13)$$

on 12,288 pooled training tokens ($\lambda = 10^{-3}$), and measured the *aligned cosine*: the mean cosine between $k_g R_{g \rightarrow h}$ and the true k_h , on 4,096 held-out tokens.

Finding 5 (heads are linearly redundant). The aligned K-cosine averages **0.962** across all layers and head pairs, with **99.75%** of pairs at or above 0.80 (and 100% above 0.60), against 0.000% under raw cosine. The apparent orthogonality of Finding 1 is a *basis artifact*: almost any head’s keys can be predicted almost perfectly as a fixed linear function of almost any other head’s keys. (A caveat: part of this predictability

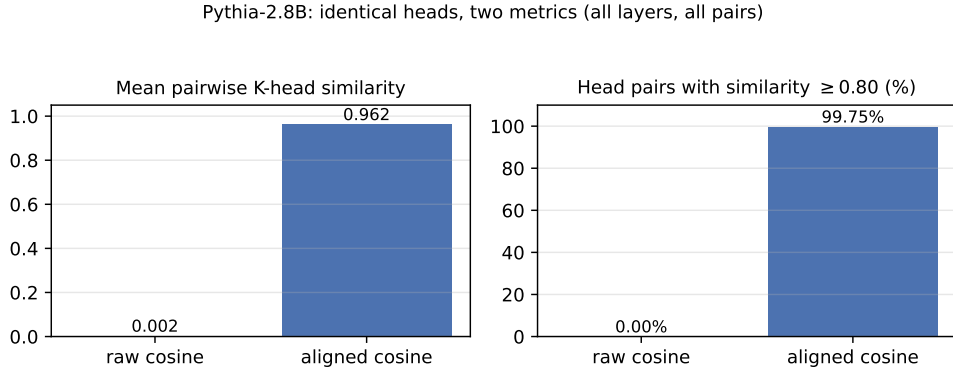


Figure 1: The measurement, not the model: the same Pythia-2.8B heads that appear orthogonal under raw cosine (left of each panel) are near-duplicates after fitting a fixed linear map per pair (right), with 99.75% of pairs above the 0.80 threshold.

Domain	G	Compr.	PPL		top-1 agreement	
			aligned	naive	aligned	naive
prose	16	2×	184	497	0.263	0.190
math	16	2×	50	320	0.378	0.172
code	16	2×	35	499	0.361	0.062
stories	16	2×	33	98	0.417	0.261
math	8	4×	116	772	0.241	0.120
code	8	4×	176	1454	0.152	0.039

Table 3: Aligned sharing on Pythia-2.8B: the group leader’s K/V is kept and member heads are reconstructed through their fitted adapters ($k_h := k_{\text{leader}}R$, $v_h := v_{\text{leader}}M$), vs. naive mean merging at the same similarity-based clustering. Baselines: prose 10.1, math 6.4, code 3.8, stories 5.1. Aligned sharing wins all 16 comparisons, by up to 14× in perplexity (code, $G=16$).

may reflect the low effective rank of the shared hidden state rather than pairwise functional identity; the functional test below is therefore the decisive one.) This phenomenon is independently documented at scale by concurrent work (Shaikh et al., 2026), and alignment-before-merging is used for offline MHA→GQA conversion by Jin et al. (2024); we regard Finding 5 as an independent confirmation, with the added control that the same clustering, budget, and evaluation are used with and without alignment.

Finding 5b (alignment generalizes to GQA models). Repeating the alignment experiment on Qwen3-4B’s 8 *uptrained* KV heads yields mean aligned K-cosine **0.893** with 95.7% of pairs ≥ 0.80 (against 0.000% raw), GQA uptraining does not remove inter-head linear structure. The functional gains are the largest in the study (Figure 3): at 2× compression beyond the model’s existing GQA ($G=4$), aligned sharing degrades perplexity by as little as +148% (stories: 12.5 vs. baseline 5.1, top-1 agreement 0.58) where naive merging degrades by +4,639%; at 4× ($G=2$), aligned sharing beats naive merging by up to **403×** (code: 32 vs. 12,985). Every one of the 16 Qwen comparisons favors aligned sharing.

Finding 5c (a third architecture, and near-usable quality). On Nemotron-Mini-4B ($L=32$, $H_q=24$, $H_{kv}=8$; a third architecture family), aligned K-cosine averages 0.932 with **100%** of pairs ≥ 0.80 , and training-free aligned sharing reaches its best quality yet: at 2× compression beyond the model’s GQA ($G=4$), code degrades by only +**34%** perplexity (4.31 vs. baseline 3.20) with top-1 agreement 0.765, stories by +65%, math by +93%, versus +3,792% to +7,257% for naive merging at the same budget. All 16 Nemotron comparisons favor aligned sharing. Across the three models the pattern is monotone in alignment quality: higher aligned cosine predicts smaller functional loss, exactly as Theorem 5.4 prescribes.

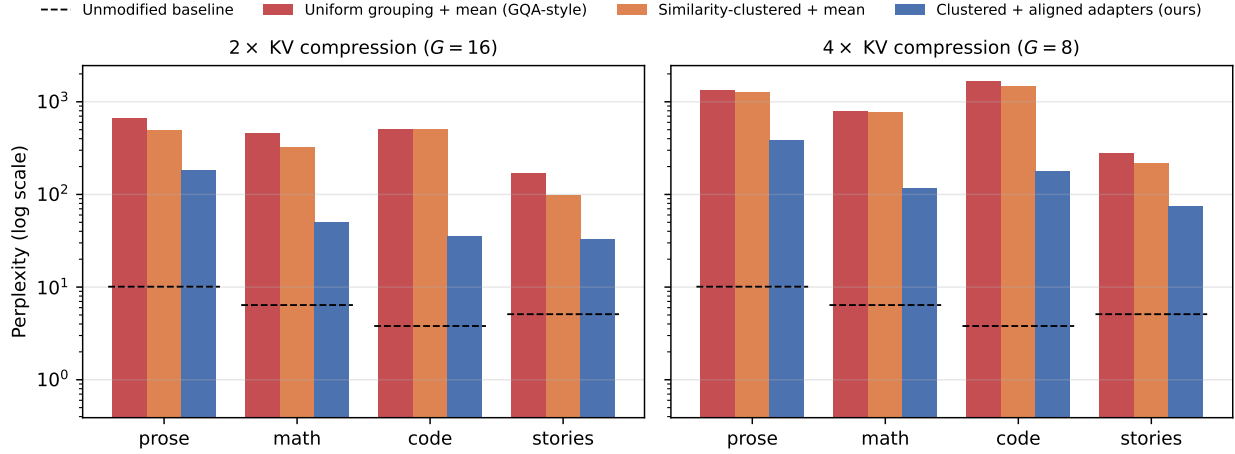


Figure 2: Benchmark on Pythia-2.8B at matched KV-cache budgets (training-free, log scale; dashes mark the unmodified baseline). GQA-style uniform grouping with mean merging (red) vs. similarity-clustered mean merging (orange) vs. our clustered aligned-adaptor sharing (blue). Ours is best in all 8 domain/budget cells, by up to 14×; the residual gap to baseline is what Stage-4 uptraining must close.



Figure 3: Benchmark on Qwen3-4B at matched budgets *beyond* its built-in GQA (training-free, log scale; dashes mark the unmodified baseline). Uniform-grouping bars are from the Stage-3 run (8K eval tokens); clustered bars from the alignment run (4K eval tokens). Aligned-adaptor sharing (blue) wins every cell, by up to 403× over naive merging at $G=2$ code.

Finding 6 (aligned sharing is far better, but not yet free). Table 3 repeats the Stage-3 simulation with adapter-reconstructed sharing. Aligned sharing beats naive merging in all 16 comparisons, by up to 14× in perplexity and 6× in top-1 agreement (code, $G=16$), and shrinks the best-case degradation at 2× compression from +1,816% to +545% (stories). The remaining gap to baseline is the practical measure of Theorem 5.4: even a 0.96 mean aligned cosine leaves a residual $\|K_h - \tilde{K}_h\|_2$ that attention logits amplify. The per-head adapters here are exactly the training-free limit of the learned shared projections of Definition 4.1; closing the residual gap is what template training (Stage 4, LoRA-scale) is for. Notably, the adapter arithmetic is cheap: reconstructing a head costs one $d_h \times d_h$ product per token ($Hd_h/d = 1/32$ of the projection FLOPs), while the cache stores only G leader heads, so the mechanism retains the full H/G cache compression of Proposition 5.1.

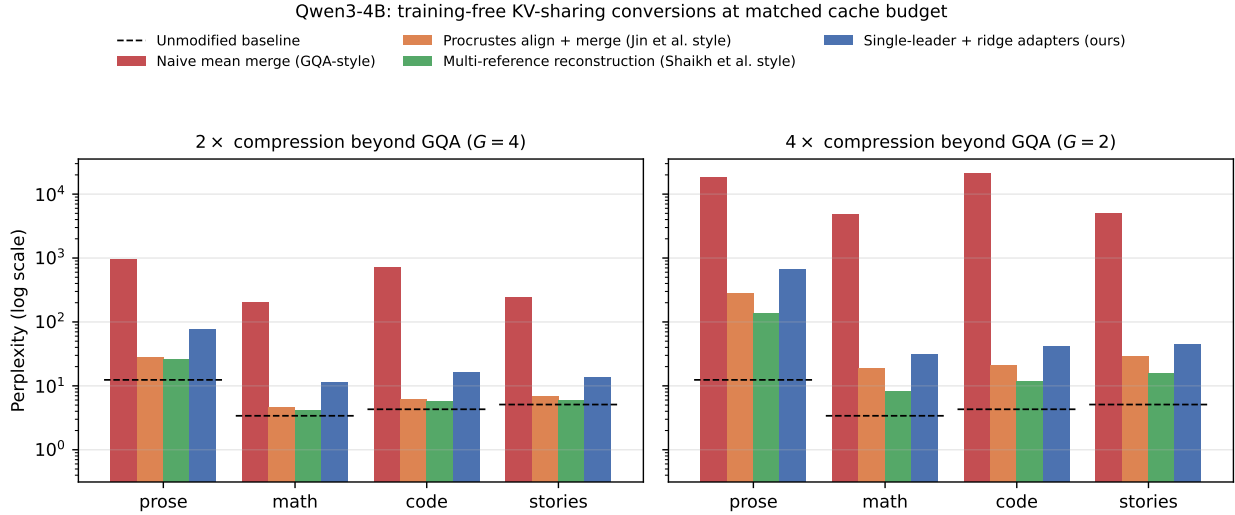


Figure 4: Head-to-head benchmark of training-free KV-sharing conversions on Qwen3-4B at matched cache budget (log scale; dashes mark the unmodified baseline). Every alignment-based method beats naive merging by one to three orders of magnitude; among them, multi-reference reconstruction is best in all 8 cells, reaching +17–20% perplexity at 2× compression (stories, math) with top-1 agreement up to 0.79.

7.4 Comparison with prior training-free conversions

Finally, we benchmark the alignment family head-to-head. On Qwen3-4B, with identical clustering, identical fit data (12,288 tokens), identical evaluation, and an identical cache budget of G stored heads per layer, we compare four training-free conversions: naive mean merging (the GQA-style conversion baseline), orthogonal-Procrustes alignment with merged aligned heads in the spirit of Jin et al. (2024), multi-reference reconstruction of each eliminated head from all G stored leaders in the spirit of Shaikh et al. (2026), and our single-leader ridge adapters. The latter two prior methods are our reimplementations of the cited ideas in this training-free, matched-budget setting, not the authors’ full pipelines (which include uptraining and per-head reference selection respectively).

Finding 7 (richer reconstruction wins; the benchmark, not our variant, is the contribution).

Figure 4 shows a consistent ordering in all 8 domain/budget cells: multi-reference > Procrustes-merge > single-leader adapters $\gg$ naive merging. At 2× compression beyond Qwen3-4B’s built-in GQA, multi-reference reconstruction costs only +17% (stories), +20% (math), +32% (code), and +111% (prose) perplexity, and at 4× it remains 2–3 orders of magnitude better than naive merging. The ordering is informative: reconstruction quality tracks the information available to the reconstructor (all G leaders > merged aligned average > one leader), exactly as Theorem 5.4 prescribes, since richer predictors shrink $\|K_h - \bar{K}_h\|_2$. All three alignment methods are instantiations of the learned-shared-projection principle of Definition 4.1; we therefore recommend the multi-reference variant as the operating point, at a modest additional adapter cost (Gd_h^2 versus d_h^2 per reconstructed head, still $\ll$ the projection FLOPs). To our knowledge this is the first controlled comparison of these conversions at strictly matched budget and data.

7.5 Verdict on the hypothesis

The strong hypothesis of Section 1: that a lightweight predictor can find input-dependent, high-similarity head clusters whose KV computation can be shared nearly for free, is *not supported* in its original, raw-activation form: the input-dependent component is weak (Finding 2) and raw similarity is absent (Finding 1). But Section 7.3 shows the redundancy the hypothesis needs *does exist* once measured correctly: heads are 0.96-similar up to fixed linear maps (Finding 5), and exploiting those maps recovers an order of magnitude of the quality lost to naive sharing (Finding 6). Three claims now carry direct empirical support: (i) in the MHA

setting, similarity-clustered groups dominate the uniform contiguous grouping that GQA implementations default to (Finding 4); (ii) KV-head redundancy is real but basis-obscured, so any viable sharing scheme must include per-head linear corrections, exactly the learned shared projections of Definition 4.1, of which our fitted adapters are the training-free limit; and (iii) residual redundancy also lives in query-side attention patterns (Finding 3). What remains unsupported is *token-level input-dependence*: alignment structure, like similarity structure, is a property of the model, so routing, if any, belongs at coarse granularity.

8 Discussion and Limitations

Where the savings are (and are not). Like GQA, Dynamic GQA reduces KV-cache memory and K/V projection FLOPs but not attention-score FLOPs (Section 3); its end-to-end speedup therefore comes primarily from relieved memory bandwidth at decode time, which is the binding constraint in practice. Aligned sharing (Section 7.3) preserves the full H/G cache compression: only leader heads are stored, and member heads are reconstructed at read time by a $d_h \times d_h$ adapter costing $d_h/d \approx 3\%$ of the projection FLOPs per reconstructed head.

Systems cost of dynamism. Token-level routing produces a ragged cache and index-gather traffic (Proposition 4.6); segment-level routing is the recommended operating point. Batched serving adds a further constraint: sequences in a batch may select different templates, which fragments batched GEMMs; per-sequence routing amortizes this.

Parameter cost. A template family with M templates of G groups carries $2MGdd_h$ KV-projection parameters versus $2Gdd_h$ for GQA. For small M (4–16) this is a modest fraction of total parameters, and templates may share a low-rank basis to reduce it further; we leave this to future work. Aligned sharing adds $2(H - G)d_h^2$ adapter parameters per layer, for Pythia-2.8B at $G=16$, about 0.4% of the attention parameters.

Relation to trained-from-scratch dynamism. DGQA (Khan et al., 2024) adapts groupings during training and fixes them at inference; mixSGA-style token routing over group sizes and MLA’s latent compression (DeepSeek-AI, 2024) are complementary axes. The aligned sharing of Section 7.3 is closest in spirit to DHA’s head fusion (Chen et al., 2024), but requires no gradient steps: adapters are closed-form ridge solutions fitted on 12K tokens. A rigorous novelty claim for a full paper must include head-to-head comparisons at matched cache budgets against GQA, QCQA, DHA, and MLA.

Limitations of the experiments. Four caveats bound the conclusions of Section 7. First, the aligned-sharing adapters are fitted, not trained end-to-end; gradient training of the same parameterization (Stage 4) should only improve on them, but this is untested. Second, part of the high aligned cosine may reflect low effective rank of the shared hidden state rather than pairwise functional identity; the functional simulation partially controls for this, since sharing quality tracks the aligned similarity structure. Third, the token budget (6–16K tokens) and model count (two) are small; the domain-stability finding (ARI) deserves replication at scale. Fourth, the theoretical results in Section 5 are soundness and accounting results, not existence results (Remark 5.8); the experiments resolve the existence question negatively for raw activations and positively for linearly-aligned activations.

9 Conclusion

We formalized Dynamic Grouped-Query Attention: input-dependent, predictive sharing of key–value computation across attention heads, routed over a small learned family of collaboration templates. The mechanism strictly generalizes GQA, provably preserves cache consistency under causal decoding, saves projection compute and cache memory in exact, quantified amounts, and introduces output error that is controlled, and vanishes under exact head redundancy, by a spectral perturbation bound. We then put the central empirical premise to a pre-registered, training-free test on Qwen3-4B and Pythia-2.8B for under US\$2 of total compute. The strong premise fails in its raw form: KV-head activations are nearly orthogonal, and their similarity structure is a property of the model far more than of the input, leaving little for a token-level router to

exploit. What survives is sharper and, we believe, more useful. First, grouping quality matters enormously in the MHA setting: similarity-clustered grouping dominates standard GQA’s uniform contiguous grouping in 11 of 12 comparisons. Second, obtained by reverse-engineering the initial negative result, and independently corroborating concurrent reports of inter-head linear structure: the redundancy the hypothesis needs does exist, hidden by a change of basis: fitting a fixed linear map per head pair raises measured similarity from 0.002 to 0.962 (99.75% of pairs above 0.80), and sharing a single leader head’s K/V through such fitted adapters, at full H/G cache compression and $\sim 3\%$ FLOP overhead, beats naive merging in all 16 comparisons by up to $14\times$ in perplexity. The theory and measurements together identify the viable form of the original idea: similarity-clustered KV grouping with per-head linear corrections, trained shared projections being their strictly more expressive counterpart, with dynamism, if anywhere, at coarse granularity or on the query side. A controlled benchmark against training-free reimplementations of prior conversions (Procrustes merging, multi-reference reconstruction) shows that reconstruction richness, not the specific variant, drives quality; the multi-reference operating point reaches $+17\text{--}20\%$ perplexity at $2\times$ compression with no training. Closing the remaining gap to full quality is precisely the Stage-4 uptraining experiment, which we leave as the immediate next step. All reported measurements were produced end-to-end on rented cloud GPUs (RTX 3090/4090 and an NVIDIA A100-SXM), for a total compute cost of under US\$5.

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